MIL-A-15199B(SHIPS)
21 May 1959
SUPERSEDING
MIL-A-15199A(SHIPS)
19 February 1952

MILITARY SPECIFICATION

ADHESIVE, ASBESTOS CLOTH TO PIPE, INSULATION

1. SCOPE

1.1 This specification covers fibrous insulation adhesive cement for securing woven asbestos cloth to insulating material employed on piping or other applications for Naval ships.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
SS-C-466 - Cloth, Thread, and Tape - Asbestos.
TT-E-485 - Enamel, Semi-Gloss, Rust-resisting.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-760 - Drums and Pails, Metal (S and 16.64 Gallon).

MILITARY
MIL-C-2861 - Cement, Insulation, High-Temperature.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue, at 33rd Street, New York 16, N.Y.)

FSC 8040

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:53Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. - Fibrous adhesive cement shall be composed of a mixture of sodium silicate and short asbestos fibers thoroughly mixed to obtain uniform distribution of the ingredients. The cement shall not contain any other materials which would have a deleterious effect upon metal, asbestos lagging and thermal insulation to which applied.

3.1.1 Cement, in the condition in which delivered, shall be suitable and effective for the purpose intended without heating or additon of other ingredients. The cement shall not deteriorate when stored for a period of one year in airtight containers.

3.2. Adhesive strength. - The adhesive strength of the cement shall be such that, when tested by the method specified in 4.5.1, a separation of the applied cloth from the insulating material will result in rupture of the insulating material and not of the cement.

3.3 When tested by the method specified in 4.5.2, the cement shall not give off objectionable or irritating fumes at temperatures up to 200° Fahrenheit (F.).

3.4 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Lot. - For purposes of sampling, a lot shall consist of all cement from one production batch offered for delivery at one time.

4.3 Sampling procedure lot acceptance (at place of manufacture). -

4.3.1 Sampling for examination of filled containers. - A random sample of filled containers shall be selected by the inspector in accordance with Standard MIL-STD-105 at inspection level I and Acceptable Quality Level = 2.5 percent defective for the examination specified in 4.4.1.

4.3.2 Sampling for acceptance tests. - From each inspection lot, two containers shall be selected at random for the tests specified in 4.4.2.

4.4 Acceptance inspection. -

4.4.1 Examination. - Each sample filled container shall be examined to verify conformance with the requirements of this specification. Any container in the sample having one or more defects or under required fill shall be rejected and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for examination provided the contractor has removed or repaired all nonconforming containers.

4.4.2 Acceptance tests. - The samples selected in accordance with 4.3.2 shall be subjected to the tests specified in 4.5. If the sample fails one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:53Z
Check the source to verify that this is the current version before use.

4.5 Test procedures. -

4.5.1 Adhesive strength. - High-temperature insulation cement conforming to Specification MIL-C-2861, shall be applied between the fixed flanges on a section of pipe of the size shown on figure 1, so that the cement will be flush with the flanges. After this material has become dry, the fibrous cement under test shall be used to apply to the surface of this insulating material, three strips of grade A asbestos cloth, conforming to Specification SS-C-466. Each strip shall be 3 inches wide by 17 inches long and shall be applied to encircle the insulation material, except that the surplus length of asbestos cloth strips shall be left free instead of lapped over to make a seam. The fibrous cement shall be allowed to dry for 72 hours at room temperature. The pipe section shall then be mounted at the uncovered ends on frictionless rollers so that the pipe will be free to turn about its longitudinal axis. Load shall be applied vertically downward on the free end of each strip of cloth, successively, until the strips are separated from the insulating material.

4.5.2 A steel plate, 12 inches square and 1/8 inch thick, shall be heavily coated on one side with the test cement. After drying for 7 days at 70°F. the plate shall be placed in an oven and heated to a temperature of 200°F. for 2 hours. No objectionable or irritating fumes shall be given off.

4.6 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level A. - The adhesive shall be furnished in 1-gallon cans or 5-gallon pails as specified (see 6.1).

5.1.1.1 One gallon cans. - The one gallon cans shall conform to type V, class 2 of Specification PPP-C-96. Exterior plan B coating and side seam striping shall be required. Cans shall be provided with wire handles which shall be galvanized or protectively coated to resist corrosion.

5.1.1.2 Five gallon pails. - The five gallon pails shall conform to type II of Specification PPP-D-760. Exterior coating of pails shall be required and shall conform to Specification TT-E-485, color olive drab. Wire handles or bails shall be galvanized or protectively coated to resist corrosion.

5.1.2 Level C. - The adhesive shall be furnished in commercial containers normally used for that purpose.

5.2 Packing. -

5.2.1 Level A. -

5.2.1.1 The 1-gallon cans shall be packed in accordance with the overseas requirements of the appendix to Specification PPP-C-96.

5.2.1.2 The 5-gallon lug cover drums shall require no packing.

5.2.2 Level B. -

5.2.2.1 The 1-gallon cans shall be packed in accordance with the domestic shipment requirements of the appendix to Specification PPP-C-96.

5.2.2.2 The 5-gallon lug cover drums shall require no packing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:53Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. - The adhesive shall be packed in a manner which will insure acceptance by common carrier and safe delivery at destination. Containers or method of shipment shall comply to the carriers rules or regulations as applicable to the mode of transportation.

5.3 Marking. - In addition to any marking required by the contract or order, interior and exterior containers shall be marked in accordance with Standard MIL-STD-129 and the requirements of the Interstate Commerce Commission Regulations when applicable.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Fibrous insulation adhesive cement should not be applied directly on any painted metal surface or fibrous glass lagging for the reason that the composition of this cement is chemically active on these materials.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships
(Project 8040-0004Sh)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:53Z
Check the source to verify that this is the current version before use.

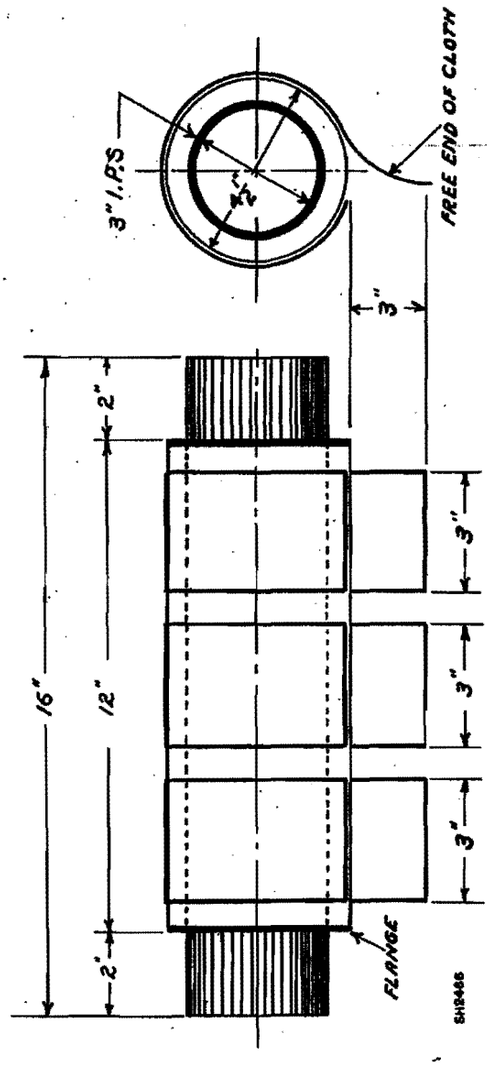


Figure 1 - Adhesive test apparatus.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:53Z
Check the source to verify that this is the current version before use.